



CSC (PPSC) 1060 1H1

Computer Science I: Java

Fall 2026

Instructor Information

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Division Phone Number
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Course Information

Course Description: Introduces students to the discipline of computer science and programming. Algorithm development, data representation, logical expressions, sub-programs and input/output operations using a high-level programming language are covered. Intensive lab work outside of class time is required.

Credit Hours: 4

Course Attribute: ASE,CTE,MED

Course Start Date: 17-AUG-26 & 17-AUG-26

Course End Date: 06-DEC-26 & 06-DEC-26

Delivery Method: Hybrid

Contact Hours: 75

Contact Hours Lecture: 45

Contact Hours Lab: 30

Class Location: PPSC Centennial Campus, Aspen Building, 250

Class Meeting Days and Times: Mon 05:30PM-07:50PM &

Drop Date: 02-SEP-26

Withdraw Date: 12-NOV-26

Note on Non-Participation: You must complete at least one academic activity in this course by 11:59 PM on the day before the drop date. If you do not participate by the deadline, you may be dropped from the class for non-participation.

Important Dates

Should include important dates (tests, capstone or final projects due, etc) but this section is optional

Course Time Commitment

General time commitment is 1 to 4 hours per credit hour (including time in class). For a 3 credit hour class, you should expect to spend between 3 and 12 hours (including time in class). For a 4 credit hour class, you should expect 4 to 16 hours. If the class has a lab associated with it, add an additional 2 to 4 hours in addition to the base commitment.

Course Outcomes

Required Course Learning Outcomes:

- 1. Demonstrate the program development process and algorithm development.**
- 2. Convert decimal numbers to and from binary and hexadecimal.**
- 3. Develop programs with thorough consideration of the Software Development Life Cycle (SDLC).**
- 4. Write programs with correct language syntax.**
- 5. Write programs with input/output from the keyboard and file using multiple data types.**
- 6. Demonstrate how operators work with different data types.**
- 7. Identify how data is represented in a computer system.**

8. Use logical expressions in a program.
9. Describe scope and lifetime rules.
10. Create programs with multiple decisions and loops.
11. Explain the concept of program flow.
12. Use system and programmer-defined functions and methods with value parameters and/or reference parameters in a program.
13. Describe recursive functions.
14. Differentiate data types in a structure and class.
15. Use pointers and references in a program.
16. Write a program with arrays.
17. Explain object-oriented methodology in program design.
18. Define the object-oriented principles.

Topical Outline

Required Topical Outline:

- I. Program development process and algorithm development
- II. Software Development Life Cycle (SDLC)
 - A. Analysis
 - B. Design
 - C. Testing
 - D. Coding standards
 - E. Documentation
- III. Number conversions
 - A. Decimal
 - B. Binary
 - C. Hexadecimal
- IV. Programming language syntax
- V. Data types and operators
- VI. Data representation
- VII. Input/Output from a keyboard
- VIII. Logical expressions
- IX. Scope and lifetime rules

- X. Decisions and iterations**
- XI. Functions and methods**
- XII. Arrays**
- XIII. Input/Output from a file**
- XIV. Pointers and references**
- XV. Object-oriented methodology**
 - A. Unified Modeling Language (UML)**
 - B. Object-Oriented Analysis and Design (OOAD)**
- XVI. Object-oriented principles**
 - A. Polymorphism**
 - B. Inheritance**
 - C. Encapsulation**
 - D. Abstraction**

Module Outcomes

STANDARD COMPETENCIES:

- 1. Demonstrate an understanding of the program development process and algorithm development.**
- 2. Implement programs utilizing analysis and design, testing, coding standards and documentation.**
- 3. Write programs with correct syntax.**
- 4. Write programs with input/output using a variety of data types.**
- 5. Demonstrate the use of different data types.**
- 6. Show how operators work with different data types.**
- 7. Identify how data is represented in the system.**
- 8. Use logical expressions in a program.**

9. Show how scope/lifetime rules affect code.
10. Write programs with multiple decisions and loops.
11. Explain program flow.
12. Use both system-defined and programmer-defined functions/methods with value and reference parameters in a program.
13. Group different data types together in a structure, class or equivalent.
14. Use pointers/references in a program.
15. Write a program with arrays.
16. Demonstrate understanding and use of recursion in a program.

Course Materials Information

**eText for Introduction to Java Programming and Data Structures
(Pearson+)**
ISBN: 8220145135180
Authors: Y. Liang
Edition: 13TH
Required or Recommended?: Required

Grade Distribution

In Class Activities/Discussion Boards 545 points

Assignments 375 points

Programming Projects 715 points

Midterm Project, Final Exam and Final Project 465 points

Total Available 2100

Final grade will be taken out of 2000 points

Grading Scale

Total Points	Percent	Letter Grade
1800–2000	90–100%	A
1600–1799	80–89.9%	B
1400–1599	70–79.9%	C
1200–1399	60–69.9%	D
0–1199	Less than 60%	F

Grades will be based on these points – grades will not be rounded up.

(For example, 1796 points is 89.8% and will be a B)

Course Schedule

See D2L Course

Assignments Overview

See D2L Course Content > Table Of Contents

Instructor Policies

No Late Work Accepted except for mid-term project with 10% point reduction from total grade for each fraction of a day submitted after due date.

In Class Activities (ICA) must be completed while present in class and turned in by end of the ICA period for credit. 27.25% point if late for ICA period.

Classroom attendance is required.

The use of AI technology to complete assignments, projects, in class activities, tests, quizzes, programming, and any other components of this class is prohibited. Students are expected to complete all components of this class on their own.

Non Participation - students will be dropped if they do NOT attend a class before Non Participation cut off date.

If you have learning challenges, please work with your accommodations office to have an approved accommodation letter.

Accommodation approval are NOT retro active, therefore students needing accommodations need these approved before needing them.

Institutional Policies

Academic Honesty, Plagiarism and Student Conduct

Pikes Peak State College places the highest value on academic honesty and integrity. Students who disregard that value deprive themselves of the learning experience they have invested in and will need for future success. Therefore, you are expected to always do your own work. Academic dishonesty includes but is not limited to submitting material prepared by someone else as your own; plagiarism (passing off another person's ideas, writings, etc., as your own); failure to cite sources adequately or correctly, using electronic data sources such as the internet, phone texting, computer IM, or smart devices such as the smartwatch or AI personal assistants without express permission of the instructor; cheating on tests by using unauthorized materials; having someone else take an online test for you; sharing your work or test answers with another student to submit as their own. If you are unclear on what counts as academic dishonesty, please consult with your instructor.

If it is proven that you have been academically dishonest, the instructor will impose the appropriate penalty as indicated on the course syllabus, up to a zero grade for the test/assignment. For more severe infractions, the instructor, in consultation with the Executive Dean, may assign a failing grade for the course.

In addition, academic dishonesty violates the PPSC Student Standards of Conduct, and a report will be made to the Dean of Students for possible disciplinary action under the [Colorado Community College System Student Behavioral Expectations and Responsibilities \(BP 4-30\)](#). Disciplinary action may occur for any standards of conduct violations, including in-class behavior which disrupts the instructor's ability to teach and other students' ability to learn. Please review the [Student Behavioral Expectations and Responsibilities \(Code\)](#).

Digital Literacy Policy

All students are expected to demonstrate digital literacy skills necessary for academic and professional success. This includes the responsible use of technology systems, ethical application of artificial intelligence (AI) tools, and professional communication in digital spaces.

Digital Literacy Expectations

- Use the Learning Management System (LMS) regularly to access announcements, assignments, and grades.
- Check college email daily for official communication.
- Access the Student Portal for registration, records, and support services.
- Use library databases and other academic resources for research and citation.

Artificial Intelligence (AI) Use

Students must follow the AI Policy outlined in the Institutional Syllabus. Individual instructors or departments may set additional expectations regarding the use of generative AI tools in their courses. Students are responsible for reviewing those expectations within each course syllabus and asking their instructor for clarification as needed.

When AI use is permitted, students must disclose when and how AI tools were used using proper citation format. One example may be found at [Brown University's LibGuide](#).

Digital Conduct & Integrity

- Cite all digital and AI sources appropriately.
- Protect personal and institutional data.
- Submit original work and respect copyright.
- Communicate respectfully and professionally in all online spaces.

Technology & Support

Students are responsible for maintaining reliable access to a computer or device capable of running required software and a stable internet connection. Support is available through the [PPSC Help Desk](#), [Division of Online Learning](#), and the [24/7 Help Desk](#).

The Ethics of Artificial Intelligence

The introduction of generative artificial intelligence (AI) tools (such as ChatGPT, and Lumen) are changing the ways we work, learn, communicate, and think. This is an exciting and challenging time, with great possibilities that also carry serious risks. At PPSC, we acknowledge that there remains a great deal of uncertainty in how to approach the use of these new tools for building online courses, creating new online pedagogies, and communicating with students in higher education. As these technologies rapidly change and evolve, we recognize that we will need to adjust our procedures more quickly than we are normally comfortable with. However, if we can center our approach to the use of AI tools with PPSC's core values, then we can engage with these technologies in ways that will maximize benefits for students, faculty, and instructors while minimizing risks.

Student Use of Artificial Intelligence (AI)

This policy outlines the guidelines and ethical considerations for the use of Artificial Intelligence (AI) technologies within Pikes Peak State College. Its purpose is to ensure that AI is used in a way that enhances learning, respects privacy, maintains academic integrity, and fosters an inclusive educational environment.

Artificial Intelligence, in particular generative AI, refers to the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, problem-solving, and understanding language. Instructors may choose from amongst three policies or any combination thereof. Option 1 allows the use of generative AI in a course with proper attribution; Option 2 would allow generative AI use at specified times with proper attribution; and Option 3 does not permit the use of generative AI at all.

Absent a separate clear statement from your course instructor, the following rules will apply to the use of generative AI in Pikes Peak State College courses.

Acceptable Uses of AI

- **AI should help you think. Not think for you. Use of tools to provide ideas, help outline thoughts, brainstorm, better understand complex problems only.**
- **Engage with AI responsibly and ethically. Share the output of AI ethically and responsibly.**
- **You are 100% responsible for your final product. As the user, if AI makes a mistake, you make a mistake. This means ideas must be attributed, facts are true, and sources must be verified.**
- **The use of AI must be open and documented. You must cite your use of AI.**

Unacceptable Uses of AI

- **Academic Dishonesty: Using AI to complete assignments, projects, tests, or any other academic work without explicit permission from the instructor is prohibited.**
- **Privacy Violations: Employing AI to monitor or track individuals without their consent or in ways that infringe on their privacy rights is strictly forbidden.**
- **Bias and Discrimination: AI applications must not perpetuate or exacerbate biases or discriminatory practices. All AI tools must be regularly audited for fairness and inclusivity.**

Regular and Substantive Interaction

In September 2023 the US Department of Education clarified the difference between distance and correspondence education. A key change of that

mandate is the requirement for “regular and substantive interaction” (RSI) between students and instructors in asynchronous online courses. RSI is defined by the Department of Education as regular and meaningful communication between students and instructors, either in real-time or through asynchronous methods as detailed in [Title 34.B.VI.A §600.2](#). In distance education, this RSI should be apparent in course design.

While issues of generative Artificial Intelligence (AI) are evolving, and PPSC encourages the use of AI as a tool for faculty and instructors within the classroom, the Department of Education has determined that “interactions with artificial intelligence . . . will not meet the statutory requirements for regular and substantive interaction ([Federal Register. Office of Postsecondary Education 2020](#)).” Students can expect that their instructor(s) will continue to participate regularly and substantively in all distance education classrooms, and that AI will only be used to supplement instruction and/or to teach AI literacy skills.

Accreditation

PPSC is accredited by the [Higher Learning Commission](#).

Americans with Disabilities Act (ADA) Academic Accommodations

Any student eligible for and needing academic accommodations due to a disability is encouraged to speak with the ACCESSibility Services (AS) Team. The following link provides additional information: [PPSC ACCESSibility Services webpage](#). You can email accessibilityservices@pikespeak.edu or contact AS at 719.502.3333. If you would like ACCESSibility Services to reach out to you directly please complete this [Accessibility Contact Request Form](#).

Reasonable accommodations are determined through an interactive process that involves the student and AS specialists. We encourage students with accommodations to discuss them with their instructors as soon as they receive them. Self-advocating for reasonable

accommodations is the responsibility of the student.

AS recommends that students request for reasonable accommodations before the first week of class. But students are welcome to make accommodation requests at any time during the semester. Students and Faculty are encouraged to contact ACCESSibility Services for information and assistance at any time.

Assessment

The faculty and staff at PPSC are committed to student learning and success. Therefore, students may periodically be asked to participate in an assessment activity for their program or department, or for the college. These activities might include taking a test, providing a writing sample, or speaking on a topic. They are designed to help faculty improve programs and teaching strategies and to promote student growth. For more information, visit the [Assessment landing page](#).

Campus Closure and Alternate Instructional Delivery

Due to circumstances beyond our control (i.e., natural disasters, extreme and extended inclement weather, pandemics, etc.) the college may change modes of instructional delivery to best accommodate student learning and academic success. If, for example, on-campus classroom instruction becomes unavailable or impossible (due to circumstances beyond our control), instruction may be delivered and provided online.

Classroom Attendance Policy

Individuals not enrolled in a class are not permitted to sit in the classroom while the class is in session. Faculty members are required to take attendance and anyone not on the class list will be asked to leave the classroom. The only exception to this procedure is for specially trained interpreters necessary for disabled students. Children are not permitted in classrooms during class meeting times.

Classroom Recordings

Students Recording

Students may record a class or discussion, including in-person or online classes and discussions, if one or both of the following conditions have been met (see [System President's Procedures 19-50](#)):

- The student has an approved disability accommodation, and/or
- The student has received written permission from the person teaching the class.

Faculty/Instructors Recording

Faculty and instructors may record class sessions, including in-person or online classes and discussions. Students should be informed if classes will be recorded. Class recordings by faculty/instructors are for the use of that specific class, unless otherwise approved.

COVID Guidelines and Mask Requirements

If you are sick, please stay home and notify your instructor. As we move into a new phase for COVID-19, we still need to be careful to take care of each other and the Pikes Peak community. With that in mind, please practice proper hygiene:

- Wash your hands regularly.
- Cover coughs and sneezes.
- Face masks are optional, faculty/instructors and staff may choose to wear masks in the classroom or on campus.
- [The Centers for Disease Control](#) recommends that people with COVID symptoms, a positive test, or exposure to someone with COVID-19, should wear a mask.

- **If you are symptomatic, please stay home and get tested.**
- **Rapid tests are available at many pharmacies and retail outlets.**
- **Students: If you are sick, stay home and inform your faculty members of your need to stay home.**
- **Employees: If you are sick, stay home and inform your supervisor of your need to stay home.**
- **If you test positive for COVID, you must remain off campus for five (5) days following your positive test report. You may return to campus on the sixth (6th) day following the positive COVID test, provided you are fever-free, and your symptoms are resolving. You should continue to wear a mask around others for five (5) additional days.**

Counseling Center

The Counseling Center is here for you all year round with free mental health services. Whether you're in crisis, feeling overwhelmed, or just need someone to talk to, we're a call or visit away. Our mission is to support student success through emotional wellness. Your mental health matters year-round. To schedule an appointment, drop in or call 719-502-4782. For more information, visit the [PPSC Counseling Center website](#).

Hours: Monday- Friday, 8 am– 5 pm

Locations:

- 1. Centennial, A141**
- 2. Rampart, N107C**
- 3. Downtown, S129**
- 4. CHES, 110**

For additional telehealth services visit the [BetterMynd website](#) and call 844-287-6963 for the BetterMynd Crisis Hotline.

Course Evaluation

Students are responsible for evaluating their course and instructor using the [MyCourses/D2L platform](#). Access to the evaluation is provided through SSO (Single Sign On), via the “My Surveys” widget on any D2L course homepage. Evaluations become available shortly after the midterm point; look for announcements in your campus email containing a survey access personalized link. Faculty and administrators use the responses collected in the survey to improve classes and programs, and individual instructors use them to improve their teaching. Evaluations are anonymous. Instructors will not see the responses until they have submitted grades, and they will not be able to match responses with individual students.

Grading

See the [current PPSC Catalog](#) for important information regarding academic standards and the grading system that applies to your course(s).

Withdrawals

Drop with a refund is possible during the first 15% of the semester. An official withdrawal may also be initiated by the student through 80% of the term resulting in a grade of “W”. A “W” grade has no credit and is not computed in the GPA. If you simply stop attending without officially withdrawing, a grade based on the total points earned will be assigned to you at the end of the semester as per the grading policy listed in the syllabus. This will usually result in an F on your grade report and may not be changed to a W once issued. Consult a current class schedule or the PPSC calendar for the exact dates.

NOTE: Your instructor cannot withdraw you; timely withdrawal is a student's responsibility.

NOTE: Students using Military and Veterans education benefits must also adhere to their benefit guidelines. Military and Veteran students who are considering withdrawing should contact the MVP office immediately to

understand the potential impacts

Incomplete Grades

An incomplete will be issued only if the student has completed more than 75% of the course requirements and has an emergency that cannot be resolved prior to the end of the semester. An incomplete is rarely issued and may pose a risk to your GPA. ALL remaining work must be satisfactorily completed by the contracted date prior to the end of the next semester, or a grade of F will be issued for the course. An Incomplete (I) grade may be removed only when the remaining class objectives are completed by the date indicated on the "Incomplete Course Agreement" form or no later than the end of the next full 15-week semester.

NOTE: Active-duty Army soldiers are required to have incompletes completed within 110 days of the end of the term. The resulting grade change is made by the instructor of record and approved by the appropriate instructional division Executive Dean. Course work not completed within the allotted time will be assigned a Failing (F) grade. Students may not re-enroll in a class in which an incomplete grade is pending, since according to the College's definition of enrollment, they are still enrolled.

Grade Change Requests

A change of grade (other than from an Incomplete) is permitted only as a result of faculty/instructor or administrative error in calculating, posting, or recording a grade. A student has one full year from the time in which the grade was issued to submit a written request for a grade reevaluation to the faculty member. Any student who wishes to pursue a change of grade must exhaust the following options in sequence.

- Grade review with faculty/instructor. If no resolution is reached or satisfactory explanation given, then:
- by department if no resolution or satisfactory explanation, then:

- Review by division Executive Dean or Associate Dean. If no resolution is reached or satisfactory explanation given, then:
- Review by the Vice President for Instructional Services, or the appointed Assistant to the Vice President, for final resolution.

Student Concerns

Examples of instructional or course concerns deal with instructor behavior, class policies, and unfair expectations or demands. Any student who wishes to pursue an instructional concern must exhaust the following options in sequence.

- The student must meet with the instructor and attempt to resolve the problem. If no resolution,
- The student must state the concern in writing via [PPSC Academic Concern Form](#) and meet with the Department Chair (in the case of an instructor) or with the Executive Dean/Associate Dean (in the case of a faculty member). Departments may require specific documentation. Please contact the appropriate division. If no resolution,
- The student will meet with the Executive Dean. If the student contests the Executive Dean's decision, he/she must submit the request in writing to the Office of the Assistant to the Vice President for Instructional Services. The request should include documentation of everything that the student wants considered in the decision. The Executive Dean will also submit all written documentation and recommendations. The Vice President for Instructional Services or a designee will notify the student of the decision in writing. This decision will be final.

The Learning Commons

The mission of the Learning Commons is to promote student persistence by reinforcing the importance of supplemental support, collaborative inquiry, and independent learning. Library, Technology, and Tutoring

Services have merged to allow for increased efficiency and effectiveness in partnering with students to develop lifelong learning strategies. Students can access computers, participate in workshops, or request academic assistance from tutors, faculty, and librarians to meet the academic demands of courses. There are a variety of other services and resources that can improve the overall student experience at PPSC. For more detailed information about services, we invite you to explore the Learning Commons or visit the website: [Learning Commons | Pikes Peak State College](#) For immediate assistance, call: 502- 2400 (Library Services) or 502-3444 (Tutoring Services).

Tutoring Center

The PPSC Tutoring Center provides free, high-quality tutoring in a variety of subjects, including math, writing, science, and more. Tutoring is for all students, at any stage, and is an important way to engage more deeply in your coursework. We encourage you to come early and come often to support your academic success. Our trained peer and professional tutors are here to support your learning—whether you are working on assignments, preparing for exams, or building study and time management skills. Tutoring is available in person and online, and most services do not require an appointment. Visit the [Tutoring Center website](#) to view locations, hours, make an appointment, or become a tutor. You can also stop by your campus Tutoring Center to get started!